Rational equations

1 A rational expression is undefined when it's denominator is equal to zero.

2

a $w = -4, 0$

b $k = -5, -\frac{3}{10}$

c $z = 0, 3$

d $v = 0$

e $m = 0$

f $b = \frac{2}{7}, 0$

g $b = 2, 5, 8$

h $p = -6, -4, 9$

i $j = -5, -1, 0$

j $a = -2, -1, \pm 7$

k $q = -3, 0, 9$

l $q = -2, 0, 5, 9$

m $q = 0, 4, 7$

n $x = 6, -1, 5, -9$

3

a Yes

b Yes

c No

d No

4

a $x = -\frac{7}{3}$

b $x = -1$

c $n = -4$

d $x = -10$

e $n = 11$

f $x = -\frac{43}{10}$

g $m = -\frac{49}{44}$

h $n = -2$

i $n = 2$

j $n = 2$

k $x = -\frac{40}{11}$

l $x = \pm\sqrt{31}$

5 Step 2 is incorrect. At this step, she incorrectly added $\frac{x}{2}$ and $\frac{x}{3}$ by adding the denominators.

6

a $x \neq 1$

b None

7 $x = \frac{6bn}{an + bm}$

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8

a

i $z = -3$

ii The equation has no valid solution.

b

i $y = -11, -3$

ii $y = -11$

c

i $z = 8$

ii $z = 8$

d

i $x = -6$

ii $x = -6$

9 The equation has no valid solution.

Applications

10 7500 cranes

11 $x = 105 \text{ ft}^2$

12 $n = 4$

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